

DLR e. V. Institut für Aerodynamik und Strömungstechnik
Lilienthalplatz 7, 38108 Braunschweig

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Contact person	Dr. Sparsh Sharma
Telephone	0531 295–1530
E-mail	sparsh.sharma@dlr.de

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Submission of a reconstructed manuscript (ref. JFM-2026-1781) to the Journal of Fluid Mechanics

Dear Editor,

We are pleased to submit the manuscript entitled “Strain-coupled one-dimensional turbulence: a single-line closure for rapid distortion, its measured limits, and the consequence for leading-edge noise” by Sparsh Sharma and Alan R. Kerstein for consideration as a regular article in the *Journal of Fluid Mechanics*.

This manuscript draws on material from JFM-2026-1781, which was declined on 18 August 2026, and per the instruction in that decision we cite the number here and request that the submission be handled by the same associate editor, Professor Sutanu Sarkar. We are not resubmitting that paper. The present manuscript is a reconstruction around what the referees identified as its strongest element—the single-line closure analysis, which two referees independently singled out and one explicitly suggested rebuilding the study around—and it contains a body of work that did not exist at the original submission: an independent DNS benchmark, a corrected linear reference, a measured closure assumption, a new relaxation mechanism developed with the model’s originator (now a co-author), and the application carried through to radiated levels. We summarise below how each of the referees’ objections has been discharged, because we took every one of them as a work order rather than a disagreement.

Validation against the model itself (Refs. 1.1, 2, 3). The central spectral results are now

benchmarked against a strained-box direct numerical simulation with exact rapid-distortion companion calculations, all reduced to the same one-dimensional observables, so that model, DNS and exact linear theory meet in a three-way (and, for the relaxation mechanism, five-way) comparison on a single benchmark observable. The strain-on/strain-off construction survives only as an internal diagnostic; every headline claim now has an external referent.

The rigid-translation “linear reference” (Ref. 3.1). The referee was right, and the point is now a result of the paper rather than an error in it: we compute exact rapid-distortion theory projected onto the line and show that the projected spectra are redistributed broadband—by a factor of four across the resolved range at unit total strain—so the correct linear reference differs from rigid translation already at the linear level. Every comparison in the paper is built on the corrected reference, and the earlier claim is withdrawn.

The single-line closure “not closed”, and not exercised (Refs. 3.2, 1.2). The closure analysis is rebuilt in exactly the modest form the objection demands: the obstruction to a single-line closure is stated exactly, sharp bounds on the rapid term are derived from line data alone (the previously underived relations now derived in an appendix), and the axisymmetry hypothesis—undefined in the original submission—is defined and *measured* rather than assumed: exact for axisymmetric distortion, with its plane-strain violation quantified as a function of accumulated strain. This machinery is used throughout the paper (the validity envelope, the deficiency localisation, the uncertainty carried into the application); the full wavenumber-dependent kernel it defines is identified, on measured grounds, as the remaining step, and we say so plainly rather than claiming it.

The plane-strain behaviour of the closure (Ref. 1.3). The moment-level validation against Lee & Reynolds has been recomputed from the exact solutions (correcting, in the process, an error in our own earlier exact curve), and the one qualitative failure—the neutral-direction anisotropy—is traced to the orientation information that every single-point closure discards, with its consequence for the spectral results assessed through the measured axisymmetry violation rather than left unexamined.

The application not carried through (Refs. 1.4, 2). The manuscript now contains the mapping the referees asked for: the model’s line spectra enter Amiet’s theory through a fitted two-parameter distorted-spectrum representation whose residual is the quoted uncertainty, yielding both a geometry-free change-of-kernel result and an assembled absolute sound-pressure-level prediction at the reference operating configuration. The computed distortion lowers the predicted level by 5–20 dB relative to the frozen von Kármán input of standard practice—a term too large to neglect, and of opposite sign to the surface-blocking correction, which we state as the scope boundary.

Presentation (Refs. 2, 3.3). The manuscript was rewritten from a fresh skeleton, not edited: established material is separated from new (the model review is explicitly marked as containing nothing original and compressed to citations), Sagaut & Cambon is adopted as the rapid-distortion spine, the leading-edge distortion literature the referees named (dos Santos, Ribeiro, Piccolo) is engaged, and the main text is shorter than the rejected version while carrying substantially more content; the validation against Chen, Meneveau & Katz and the complete intervention family are provided as supplementary material.

One structural decision deserves a sentence. The new results—the closure analysis, the

mechanism study and the noise consequence—could have been split into separate papers; we judged that separated they would each lose the evidential chain that gives them force. They are therefore presented as one paper in three parts, with the supplementary material absorbing what the arc does not need.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. Both authors have approved the submission and take responsibility for the content of the manuscript. A declaration of the use of AI tools is included in the manuscript in accordance with Cambridge University Press guidance. To the best of our knowledge, there are no conflicts of interest that would affect the review of this work. We are of course content for the original referees to see how their objections were used.

Thank you for considering this manuscript for publication in the *Journal of Fluid Mechanics*.

Yours sincerely,

Dr. Sparsh Sharma, on behalf of the authors
German Aerospace Center (DLR)
Institut für Aerodynamik und Strömungstechnik
Lilienthalplatz 7
38108 Braunschweig, Germany
sparsh.sharma@dlr.de